UNIVERSITY OF OXFORD

DEPARTMENT OF PAEDIATRICS

METHODOLOGY, EVIDENCE SYNTHESIS, AND IMPLEMENTATION | MESISIG
IASP Special Interest Group

Navigating AI in Evidence Synthesis

AI tools across the review workflow

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Map and presentation

lukebax.github.io/ai-review-tools/

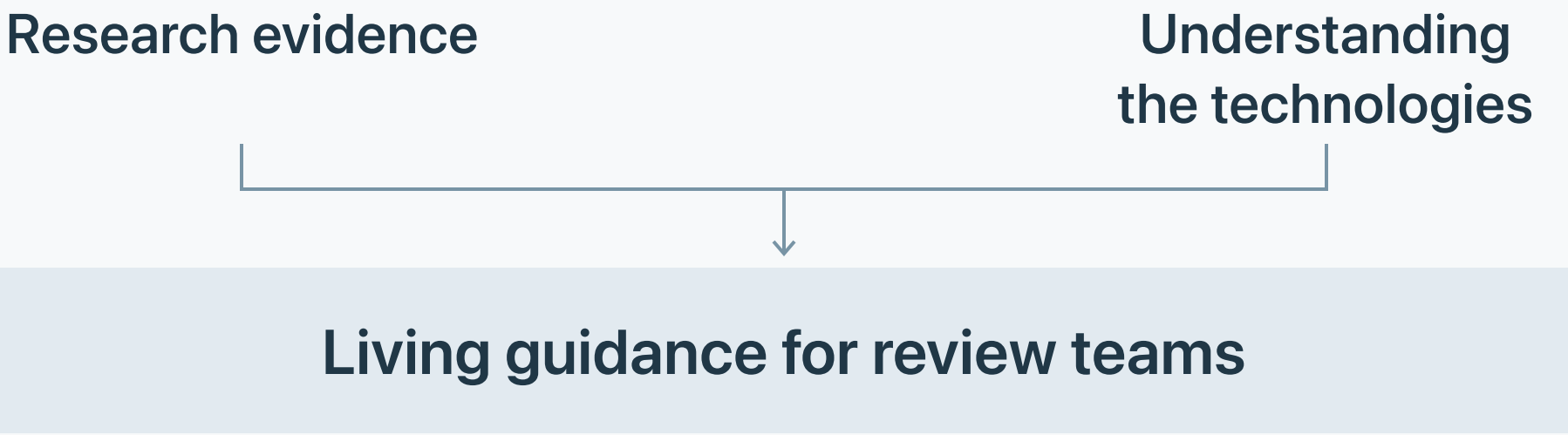
Stages of a systematic review

1 Protocol › 2 Search › 3 Screening for relevance › 4 Quality assessment

5 Data extraction › 6 Data synthesis › 7 Certainty of evidence › 8 Manuscript

RAISE provides guidance for using AI in reviews

Responsible use of AI in evidence synthesis

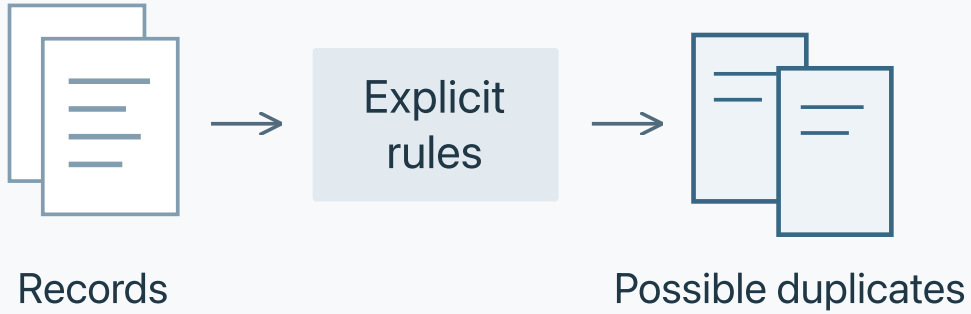


Developed by researchers working in evidence synthesis

Explicit rules and patterns in unlabelled data

Rule-based algorithms

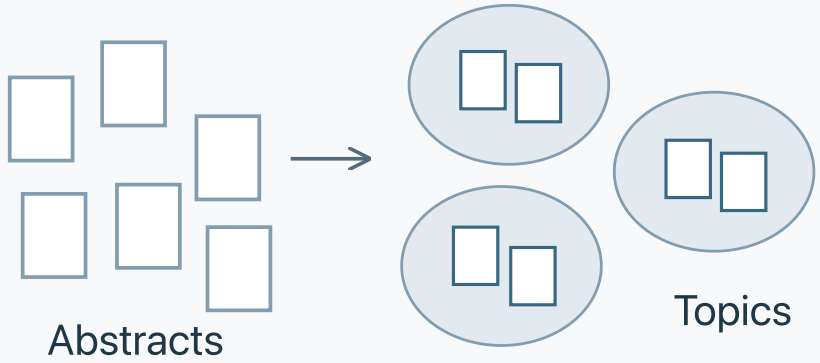
Deduplication



Predefined rules compare the records.

Unsupervised machine learning

Topic modelling



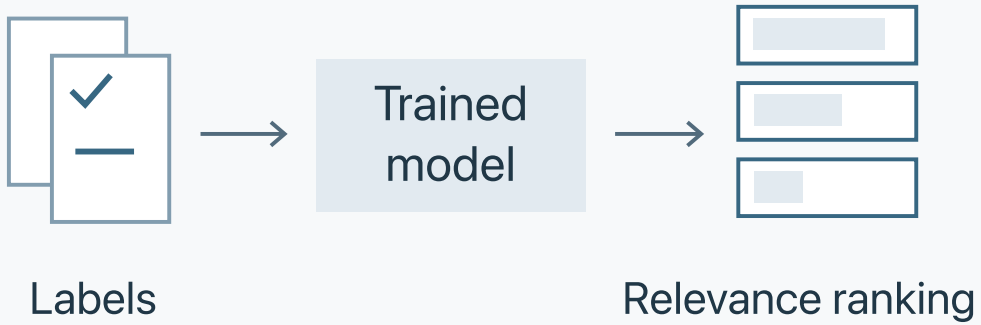
Patterns emerge without supplied labels.

These are broad, overlapping categories; individual tools can combine approaches.

Learning from labelled examples and generating language

Supervised machine learning

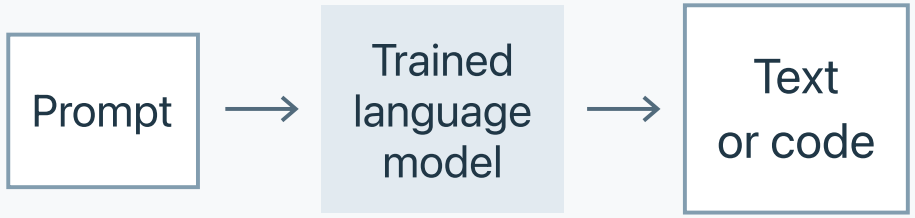
Relevance screening



Already trained or trained during the review

Generative LLMs

Writing analysis code



Already trained, with prompts tailored to the task

These are broad, overlapping categories; individual tools can combine approaches.

Recommendations apply to a particular use

Acceptable for use	Use outputs directly, accounting for limitations and bias
Human verification required	Check outputs and correct errors
Requires validation within the review	Test performance against suitable prespecified thresholds and benchmarks
Exploratory and supplementary use	Explore or supplement established methods
Not acceptable for use	Do not rely on outputs for this task

Foundations for mapping the AI tool landscape

RAISE A leading framework of guidance for responsible use of AI in evidence synthesis

Review stages

Rows

1. Writing a protocol
2. The Search
3. Screening for relevance
4. Quality assessment
5. Data extraction
6. Data synthesis
7. Certainty of evidence assessment
8. Writing a manuscript

AI categories

Column groups

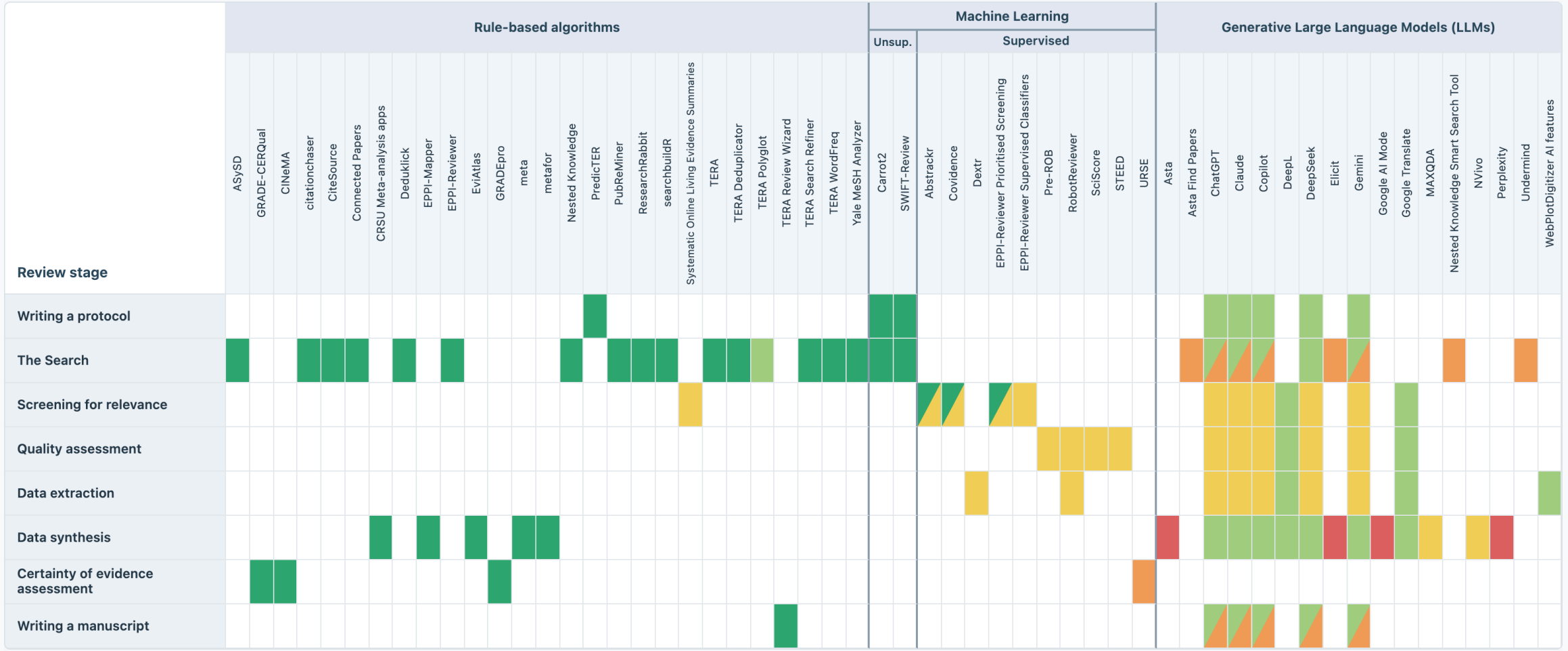
1. Rule-based algorithms
2. Unsupervised machine learning
3. Supervised machine learning
4. Generative large language models (LLMs)

Use recommendations

Cell colours

1. Acceptable for use
2. Human verification required
3. Requires validation within the review
4. Exploratory and supplementary use
5. Not acceptable for use

AI tools across the review stages



Find interactive map here: lukebax.github.io/ai-review-tools/

Reading the patterns in the map

Selected tools and tasks from RAISE's guidance

RAISE 3 v4, Table 2 · 56 tools/platforms

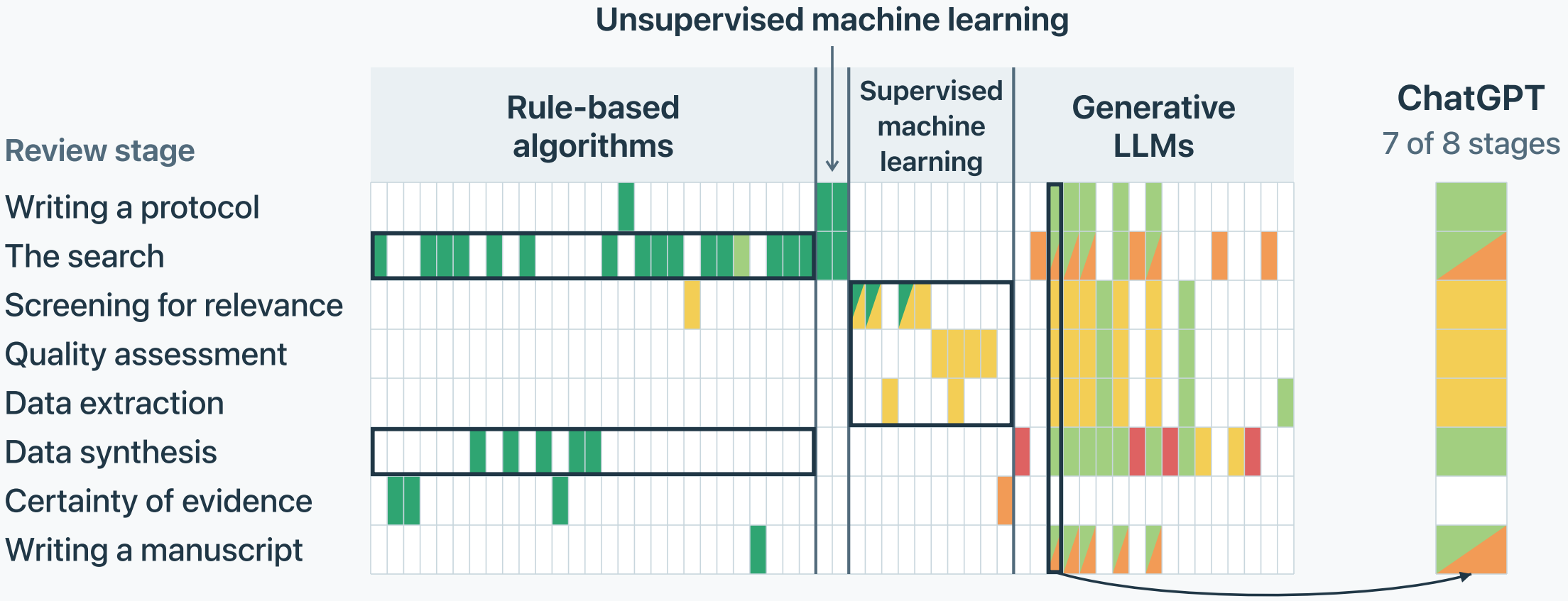
Tasks, maturity, supporting evidence and how they work all vary together

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The patterns do not independently establish that one AI family is safer.

Tools differ in where and how broadly they are used

Stage clusters in the overview, with ChatGPT enlarged alongside



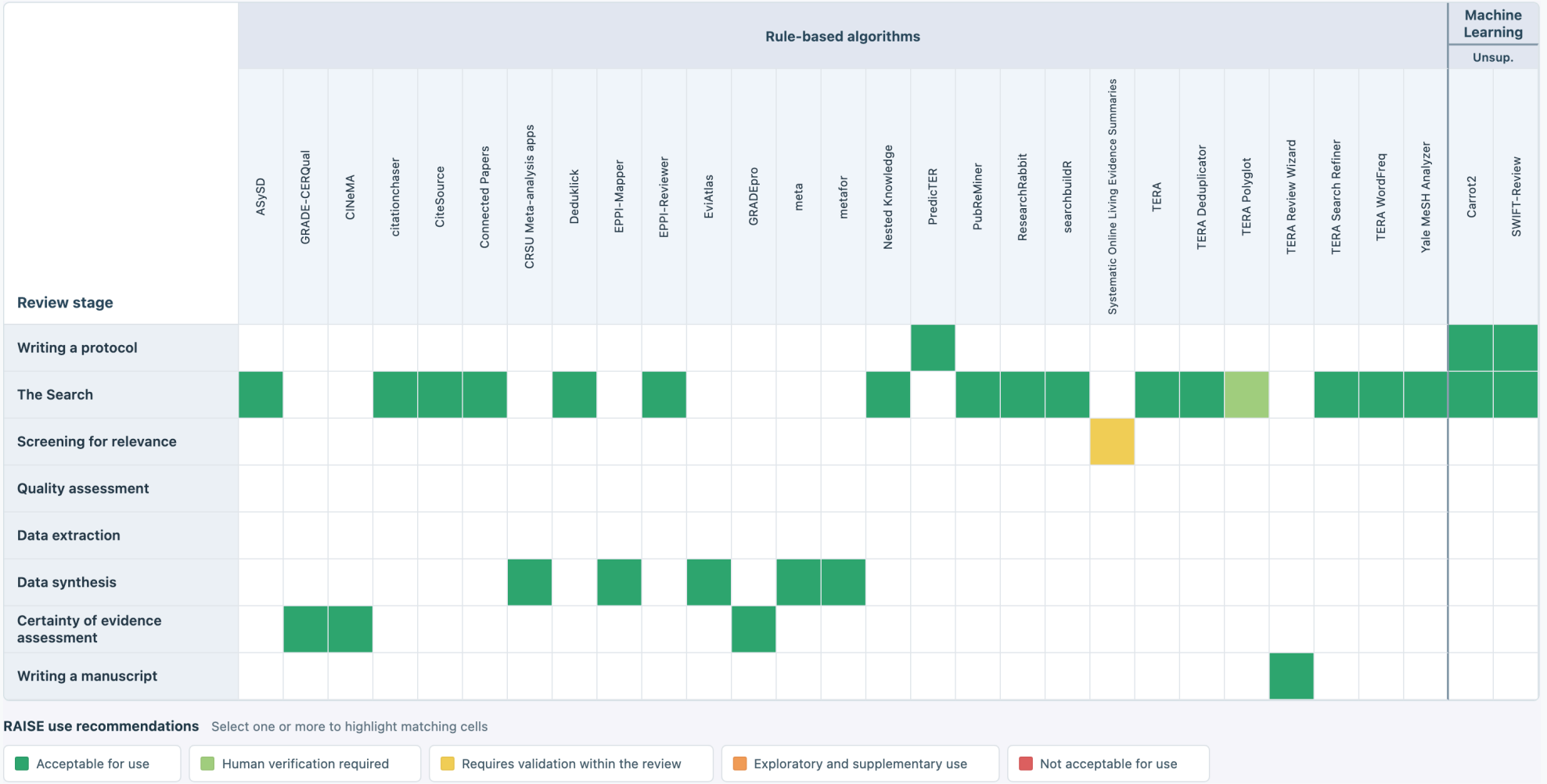
Rule-based
Record matching and calculations

Supervised
Relevance, methods and data

ChatGPT
Language and code across stages

Rule-based algorithms and unsupervised machine learning

Most of these mapped uses are acceptable



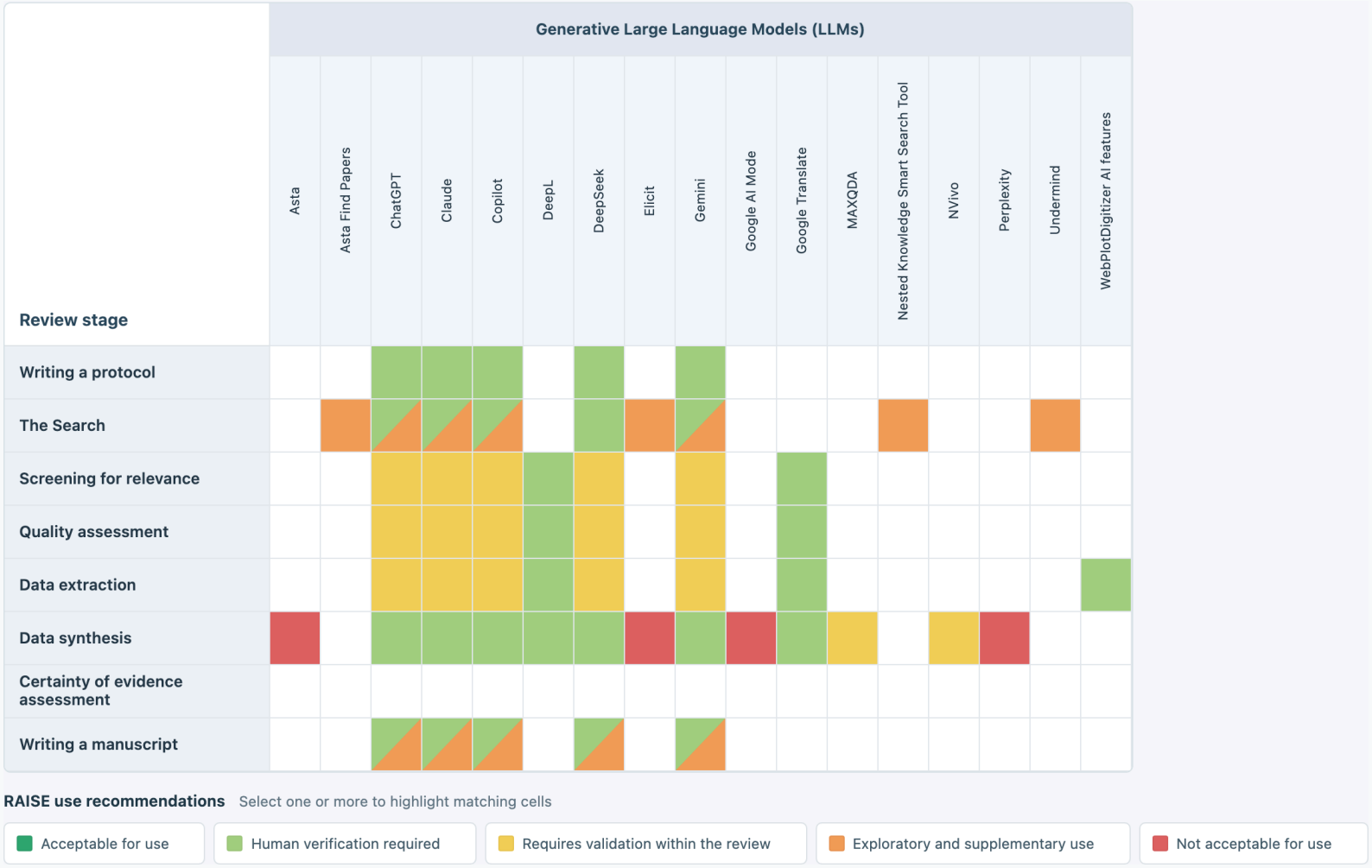
Supervised machine learning

Most mapped uses need validation: performance depends on task and material



Generative large language models (LLMs)

Recommendations depend on the task, evidence and safeguards.



Prioritised screening with and without early stopping

Stopping early changes reliance on the ranking

Abstrackr



Covidence



EPPI-Reviewer Prioritised Screening



Screen every record

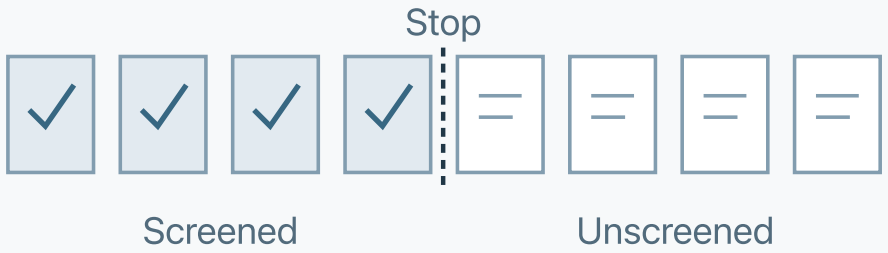


Every record is screened

Acceptable for use

A poor ranking does not leave records unseen.

Stop early



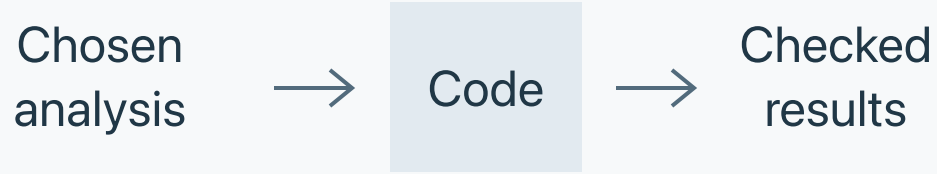
Requires validation within the review

Assess the screening and stopping procedure.

The ranking can be identical. What changes is how much we rely on it.

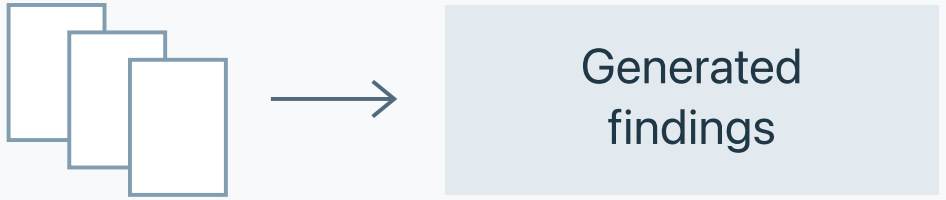
Synthesising findings differs from drafting analysis code

LLM-drafted code for a chosen analysis



Human verification required

LLM-derived synthesis across studies



Not acceptable for use in a formal review

RAISE also restricts complete agentic reviews

Complete agentic review: not acceptable for use



Using an LLM agent for one review task requires its own assessment.

Tool suitability

AI-assisted relevance screening

1

What is the purpose of the tool?

Apply our inclusion and exclusion criteria



2

Where have the training, testing and validation data come from?

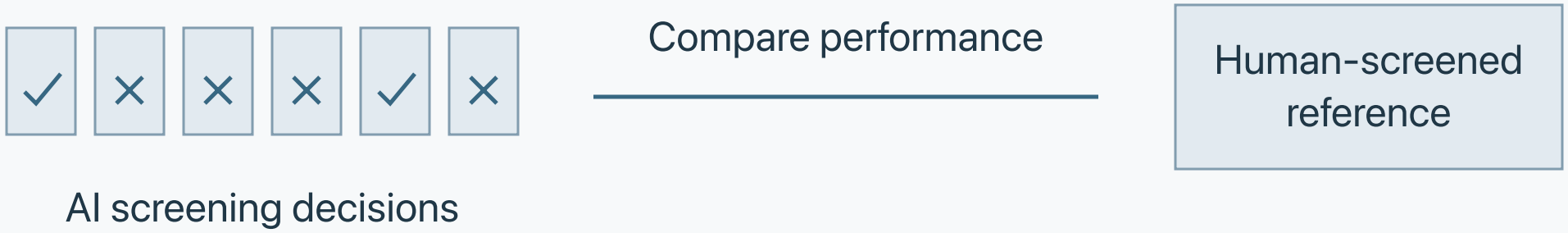
Training data: relevance and quality.
Testing and validation: independence and relevance.



Validation and performance

AI-assisted relevance screening

3 Is the tool validated and performing sufficiently well for use?



Check for missed eligible studies

An appropriate human-screened reference

Set the performance standard in advance.
Establish whether it is met within this review.

Team, resources and documentation

AI-assisted relevance screening

4 Does the team have the expertise and resource to use the AI tool effectively, and is the usability acceptable?

Skills, assessment time and a workable process

5 Is there sufficient transparency around licenses and other available documentation?

Instructions, access, data handling, version and settings

Resolve gaps in the evidence before relying on the tool’s decisions.

Choosing a suitable use of AI

Consider how the tool works, the task and the evidence needed before relying on its outputs.



RAISE provides the guidance.

The companion map helps you explore RAISE's recommendations and the tools.

Map and presentation

lukebax.github.io/ai-review-tools/